GEOMETRY · UNIT 1: FOUNDATIONS OF GEOMETRY

Treasure Map Navigator

Use distance, midpoint, and angle relationships to find buried treasure.

CP1

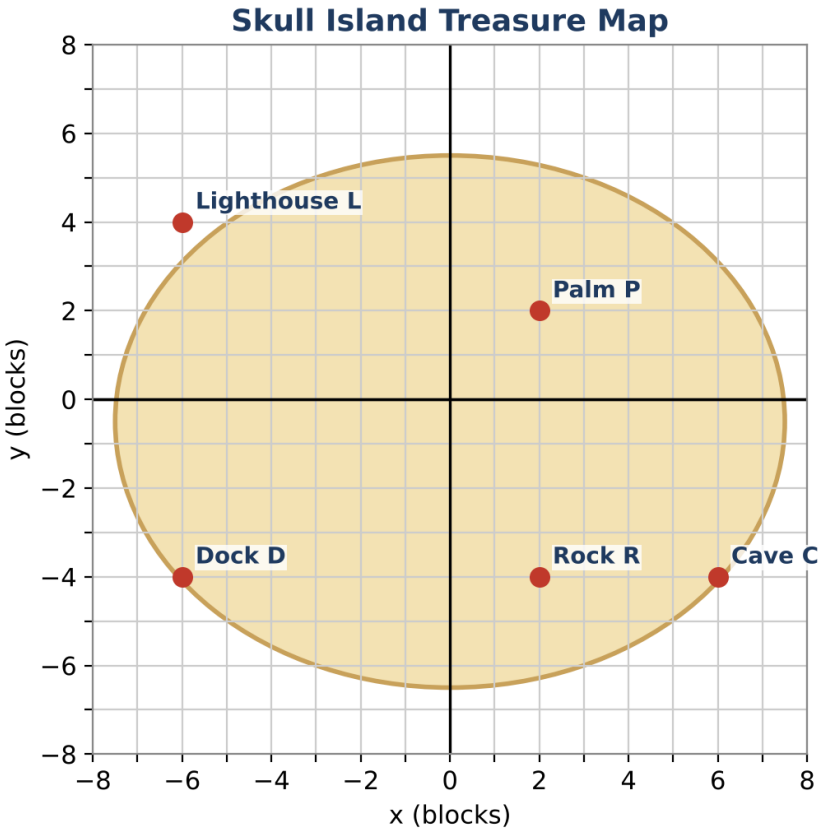
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

A pirate's map of Skull Island has turned up in the school library. The landmarks sit on a coordinate grid where each unit is one block.

Each clue uses a geometry idea: distance, midpoint, segment addition, or angle relationships. Solve the clues to find the treasure, then make a map of your own.



Learning Goals

- I can use the distance and midpoint formulas.
- I can solve for unknowns with segment addition, angle addition, and angle bisectors.
- I can use angle pair relationships to find measures.

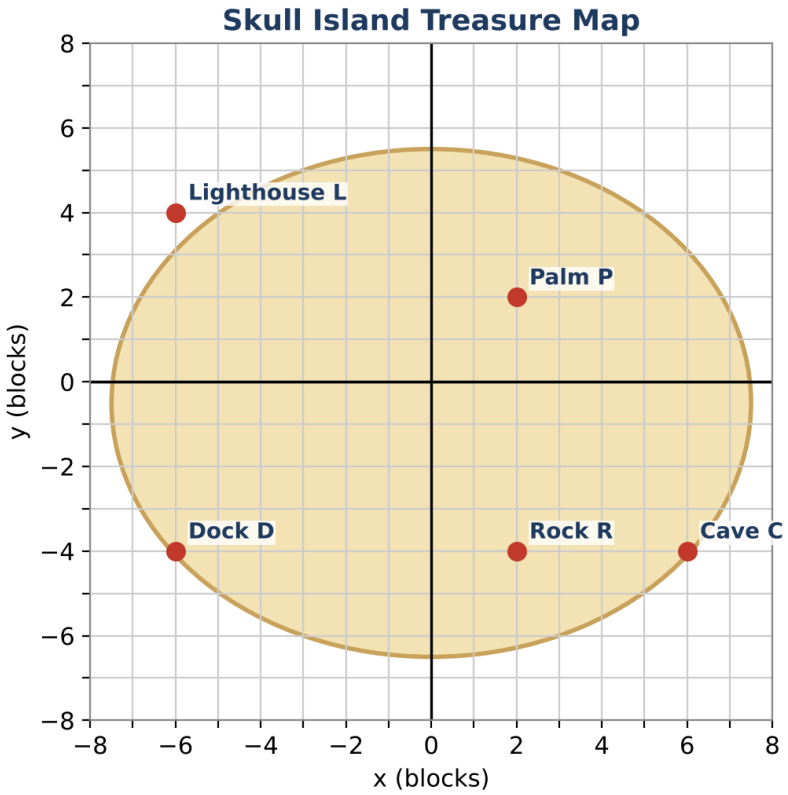
Project Timeline

When	What you will do
Day 1	Distance & Midpoint Clues
Day 2	Segment & Angle Clues
Day 3	Design Your Own Map
Final	Treasure Map & Clue Set + reflection

Part 1: Distance & Midpoint Clues

Day 1 · Use the map and the coordinate table.

1. Find DP, PC, and DC. Give exact answers and decimals to the nearest hundredth.



Location	Coordinates
Dock D	(−6, −4)
Palm P	(2, 2)
Cave C	(6, −4)
Lighthouse L	(−6, 4)
Rock R	(2, −4)

2. Find the perimeter of the triangle formed by the Dock, the Palm, and the Cave.

3. The Cave is the midpoint of the Palm and a hidden spot X. Find the coordinates of X.

4. Find the midpoint of the Lighthouse and the Cave. Which landmark is closest to it?

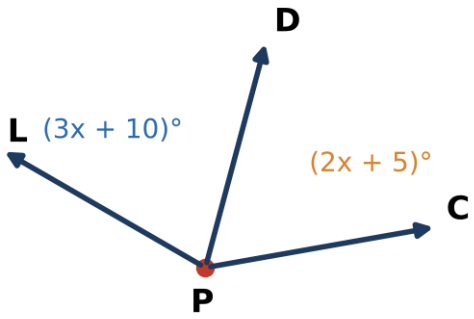
Part 2: Segment & Angle Clues

Day 2 · Write and solve an equation for each clue.

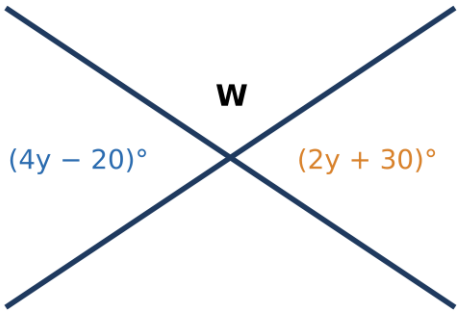
5. D, R, and C are collinear with R between D and C. DR = 2x + 2, RC = x + 1, and DC = 12. Find DR and RC. Do these match the map?

6. Ray PD bisects ∠LPC. m∠LPD = (5x − 4)° and m∠DPC = (3x + 12)°. Find m∠LPC.

Angles at the Palm (not to scale)



Paths crossing at the Well



7. Two straight paths cross at the Well. One angle is (4y − 20)° and its vertical angle is (2y + 30)°. Find all four angles formed.

8. Two angles at the Rock are complementary. One is 18° more than twice the other. Find both.

Part 3: Design Your Own Map

Day 3 · Design a map whose treasure can only be found by solving every clue.

9. Place five landmarks and write a clue that uses "the midpoint of A and B is C, find B".

10. Write a segment addition clue and an angle bisector clue with algebra.

11. Write the answer key. Trade with a partner and record any clue they found unclear.

Final Product: Treasure Map & Clue Set

Your final product must include:

- Labeled map with coordinates
- Clues using distance, midpoint (including a missing endpoint), segment addition, and an angle bisector
- Complete answer key
- Partner feedback and revision

Reflection

R1. How can you find a missing endpoint if you know the midpoint?

R2. What is the difference between a bisector and a midpoint?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Original clues correctly use distance, midpoint, and angle relationships and lead to one location. Goes beyond the minimum with a creative, realistic choice.	Original clues correctly use distance, midpoint, and angle relationships and lead to one location.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Map and clue set show labeled coordinates, every clue, and a complete answer key. Polished, organized, and visually engaging.	Map and clue set show labeled coordinates, every clue, and a complete answer key. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.